

DCI-FUSE: A Low-Rank Generalization of Unsupervised Score Ensembling under Latent Verification Regimes

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Abstract

This paper generalizes the Fully Unsupervised Score Ensembling (FUSE) framework by relaxing its implicit rank-one dependency constraint. We propose the Deterministic/Conditional Independent (DCI) generative model, which identifies that verifier dependencies are governed by multiple latent regimes. We prove that, under heterogeneous conditions, the pooled covariance matrix exhibits a fundamental low-rank structure, generalizing FUSE as the special case where $K = 1$. By deriving the tensor decomposition of third-order moments, we establish the identifiability of these latent regimes via Kruskal’s theorem. Our framework transitions score ensembling from simple rank-one spectral estimation to a robust, low-rank subspace recovery paradigm.

Summary of Contributions:

- **Generalization:** We provide a rigorous rank- K spectral framework that generalizes FUSE to handle arbitrary latent verification regimes.
- **Optimality Analysis:** We prove that FUSE, as a rank-1 estimator, inherently suffers from irreducible systemic bias when applied to heterogeneous mixtures (Theorem 5.3).
- **Identifiability:** We derive the identifiability of latent regimes through third-order tensor decomposition, ensuring precise parameter recovery via Kruskal’s theorem.

1 DCI-FUSE: Complete Theoretical Analysis

1.1 Latent Regime Framework and Structural Settings

Proposition 1.1 (Formalization of Latent Verification Regimes). *In the unsupervised ensembling literature, given a query q , the covariance matrix is typically computed across N independent responses, implicitly assuming a singular, homogeneous verifier dependency structure. Our DCI framework relaxes this by introducing a latent regime variable d , allowing for two distinct structural paradigms:*

- **Setting A (Query-level d , Latent Manifold Recovery):** *The regime d is invariant across responses to a single query but exhibits variation across queries. While a single-query covariance collapses the rank of the latent structure (due to vanishing cross-regime variance), vertically stacking score matrices across L queries into a grand matrix $\mathbf{V} \in \mathbb{R}^{(NL) \times m}$ restores the underlying low-rank structure. This approach recovers the global dependency geometry driven by cross-query regime shifts.*
- **Setting B (Response-level d , Query-conditional Mode):** *The regime d_i varies across individual responses within a single query. In this mode, the mixture properties manifest locally, allowing the low-rank structure to be captured within a single query-conditional matrix.*

This work adopts **Setting A (Regime-Agnostic Decomposition)** as its theoretical cornerstone, as it accurately models real-world evaluation scenarios where responses are sampled from a heterogeneous mixture of latent verification conditions.

1.2 Generative Model (Latent Manifold Recovery)

Consider L independent queries $q_1, \dots, q_L$, where each query contains N responses, yielding a total of NL observations.

$$d_\ell \sim \text{Cat}(\rho_1, \dots, \rho_K), \quad \rho \in \Delta^{K-1}, \quad (1)$$

$$b_\ell := b^{(d_\ell)}, \quad \text{i.e., class balance depends directly on the regime,} \quad (2)$$

$$y_{\ell i} \mid d_\ell = k \sim \text{Bern}\left(\frac{1 + b^{(k)}}{2}\right), \quad y_{\ell i} \in \{-1, +1\}, \quad (3)$$

$$v_{\ell ij} \mid y_{\ell i}, d_\ell = k \sim P_j(\cdot \mid y_{\ell i}, k). \quad (4)$$

1.3 DCI Assumption

$$P(v_{\ell i1}, \dots, v_{\ell im} \mid y_{\ell i}, d_\ell) = \prod_{j=1}^m P(v_{\ell ij} \mid y_{\ell i}, d_\ell). \quad (5)$$

1.4 Moment Variables

Define the binary-transformed verifier outputs $\tilde{v}_{\ell ij} \in \{-1, +1\}$ and the regime-dependent accuracy $\pi_j^{(k)} := P(\tilde{v}_{\ell ij} = y_{\ell i} \mid d_\ell = k)$. The centered reliability vector entry is defined as:

$$u_j^{(k)} := \sqrt{1 - (b^{(k)})^2(2\pi_j^{(k)} - 1)}. \quad (6)$$

2 Second-Order Moment Decomposition

Lemma 2.1 (Off-diagonal Signal Structure). *Let M be the population pooled covariance matrix evaluated over the entire mixture distribution under DCI. The clear algebraic signal structure holds strictly for off-diagonal entries, whereas the diagonal elements are contaminated by idiosyncratic verifier-specific variance noise:*

$$M_{ab} = \Sigma_{ab}^{pool} = \sum_{k=1}^K \rho_k u_a^{(k)} u_b^{(k)} + (\mathbf{T}_{\mathbf{II}})_{ab} \quad \text{for } a \neq b, \quad (7)$$

$$M_{aa} = \sum_{k=1}^K \rho_k (u_a^{(k)})^2 + (\mathbf{T}_{\mathbf{II}})_{aa} + \sigma_a^2, \quad (8)$$

where $\sigma_a^2 = \mathbb{E}_{d,y}[\text{Var}(\tilde{v}_a \mid y, d)]$ denotes the residual variance component, and $\mathbf{T}_{\mathbf{II}}$ represents the class-balance correction matrix defined in Theorem 2.2.

Theorem 2.2 (Pooled Covariance Structure under Heterogeneous Class Balance). *Under the generative batched DCI model, without requiring the restrictive assumption that class balance $b^{(k)}$ is uniform across different regimes, the off-diagonal block of the global pooled covariance matrix satisfies:*

$$\Sigma_{off}^{pool} = [\mathbf{T}_{\mathbf{I}} + \mathbf{T}_{\mathbf{II}}]_{off}, \quad (9)$$

where $\mathbf{T}_{\mathbf{I}} = \sum_{k=1}^K \rho_k u^{(k)} (u^{(k)})^\top$ and $\mathbf{T}_{\mathbf{II}}$ is a correction matrix representing the covariance of centered conditional expectations. Consequently, its rank is strictly bounded by:

$$\text{rank}(\Sigma_{off}^{pool}) \leq 2K - 1. \quad (10)$$

Proof. For the global pooled mixture distribution, a random observation $(\tilde{v}_{\ell i1}, \dots, \tilde{v}_{\ell im})$ marginalizes over d_ℓ as $P(\tilde{v}_{\ell i\cdot}) = \sum_{k=1}^K \rho_k \sum_{y \in \{\pm 1\}} P(y \mid k) \prod_j P(\tilde{v}_{\ell ij} \mid y, k)$. We compute $\Sigma_{j_1 j_2}^{pool} = \text{Cov}(\tilde{v}_{\ell i j_1}, \tilde{v}_{\ell i j_2})$ by applying the Law of Total Covariance, conditioning on the latent query regime d_ℓ :

$$\text{Cov}(\tilde{v}_{\ell i j_1}, \tilde{v}_{\ell i j_2}) = \underbrace{\mathbb{E}_{d_\ell} [\text{Cov}(\tilde{v}_{\ell i j_1}, \tilde{v}_{\ell i j_2} \mid d_\ell)]}_{\text{Term I}} + \underbrace{\text{Cov}_{d_\ell} (\mathbb{E}[\tilde{v}_{\ell i j_1} \mid d_\ell], \mathbb{E}[\tilde{v}_{\ell i j_2} \mid d_\ell])}_{\text{Term II}}. \quad (11)$$

Step 1: Evaluation of Term I via Internal Label Decomposition.

To compute the inner conditional covariance for a fixed regime $d_\ell = k$, we apply the Law of Total Covariance a second time, conditioning on the true response label $y_{\ell i}$:

$$\begin{aligned} \text{Cov}(\tilde{v}_{\ell i j_1}, \tilde{v}_{\ell i j_2} \mid d_\ell = k) &= \mathbb{E}_y [\text{Cov}(\tilde{v}_{\ell i j_1}, \tilde{v}_{\ell i j_2} \mid y_{\ell i}, d_\ell = k)] \\ &\quad + \text{Cov}_y (\mathbb{E}[\tilde{v}_{\ell i j_1} \mid y_{\ell i}, k], \mathbb{E}[\tilde{v}_{\ell i j_2} \mid y_{\ell i}, k]). \end{aligned} \quad (12)$$

By the DCI assumption, verifiers are strictly conditionally independent given the ground truth label $y_{\ell i}$. Therefore, for all off-diagonal indices $j_1 \neq j_2$:

$$\text{Cov}(\tilde{v}_{\ell i j_1}, \tilde{v}_{\ell i j_2} \mid y_{\ell i}, d_\ell = k) = 0,$$

which eliminates the first expectation term.

Step 2: Explicit Computation of Conditional Mean over Labels.

Expanding expectations for binary verifier spaces yields:

$$\mathbb{E}[\tilde{v}_{\ell ij} \mid y_{\ell i} = +1, d_\ell = k] = (+1)\pi_j^{(k)} + (-1)(1 - \pi_j^{(k)}) = 2\pi_j^{(k)} - 1, \quad (13)$$

$$\mathbb{E}[\tilde{v}_{\ell ij} \mid y_{\ell i} = -1, d_\ell = k] = (-1)\pi_j^{(k)} + (+1)(1 - \pi_j^{(k)}) = -(2\pi_j^{(k)} - 1). \quad (14)$$

This can be expressed compactly as $\mathbb{E}[\tilde{v}_{\ell ij} \mid y_{\ell i}, k] = y_{\ell i}(2\pi_j^{(k)} - 1)$.

Step 3: Derivation of Covariance Over y .

Using the properties of the variance operator on the Bernoulli label distribution:

$$\text{Cov}_y(\mathbb{E}[\tilde{v}_{\ell ij_1} \mid y_{\ell i}, k], \mathbb{E}[\tilde{v}_{\ell ij_2} \mid y_{\ell i}, k]) = (\mathbb{E}[y_{\ell i}^2] - (\mathbb{E}[y_{\ell i}])^2)(2\pi_{j_1}^{(k)} - 1)(2\pi_{j_2}^{(k)} - 1). \quad (15)$$

Since $y_{\ell i} \in \{-1, +1\}$, we have $y_{\ell i}^2 = 1$ deterministically. Given $\mathbb{E}[y_{\ell i} \mid d_\ell = k] = b^{(k)}$, the label variance equals $1 - (b^{(k)})^2$. Substituting this back yields:

$$\text{Cov}(\tilde{v}_{\ell ij_1}, \tilde{v}_{\ell ij_2} \mid d_\ell = k) = (1 - (b^{(k)})^2)(2\pi_{j_1}^{(k)} - 1)(2\pi_{j_2}^{(k)} - 1) = u_{j_1}^{(k)} u_{j_2}^{(k)}. \quad (16)$$

Taking the outer expectation over all regimes d_ℓ , we obtain the primary low-rank signal matrix $\mathbf{T}_\mathbf{I}$:

$$\text{Term I} = \sum_{k=1}^K \rho_k u_{j_1}^{(k)} u_{j_2}^{(k)} \implies \mathbf{T}_\mathbf{I} = \sum_{k=1}^K \rho_k u^{(k)} (u^{(k)})^\top = U \text{diag}(\rho) U^\top, \quad \text{rank}(\mathbf{T}_\mathbf{I}) \leq K. \quad (17)$$

Step 4: Evaluation of Term II (The Class Balance Correction Space).

Let $\mu_j^{(k)} := \mathbb{E}[\tilde{v}_{\ell ij} \mid d_\ell = k] = b^{(k)}(2\pi_j^{(k)} - 1)$ denote the conditional mean, and let $\bar{\mu}_j := \sum_{k=1}^K \rho_k \mu_j^{(k)}$ represent the grand marginalized pooled mean. Term II expands as:

$$\text{Term II}_{j_1 j_2} = \sum_{k=1}^K \rho_k (\mu_{j_1}^{(k)} - \bar{\mu}_{j_1})(\mu_{j_2}^{(k)} - \bar{\mu}_{j_2}). \quad (18)$$

Defining the centered conditional mean vector as $\tilde{\mu}^{(k)} = \mu^{(k)} - \bar{\mu} \in \mathbb{R}^m$, this component forms the matrix $\mathbf{T}_\mathbf{II} = \sum_{k=1}^K \rho_k \tilde{\mu}^{(k)} (\tilde{\mu}^{(k)})^\top$. Crucially, by construction, the centered vectors obey the linear constraint $\sum_{k=1}^K \rho_k \tilde{\mu}^{(k)} = \sum_{k=1}^K \rho_k (\mu^{(k)} - \bar{\mu}) = \mathbf{0}$. This linear dependence ensures that these K vectors span a subspace of dimension at most $K - 1$, meaning:

$$\text{rank}(\mathbf{T}_\mathbf{II}) \leq K - 1. \quad (19)$$

Combining both terms via the subadditivity of matrix rank yields the final bound:

$$\text{rank}(\Sigma_{\text{off}}^{\text{pool}}) \leq \text{rank}(\mathbf{T}_\mathbf{I}) + \text{rank}(\mathbf{T}_\mathbf{II}) \leq K + (K - 1) = 2K - 1. \quad (20)$$

□

Corollary 2.3 (Collapse under Uniform Class Balance). *In the special case where class balance is homogeneous across all regimes ($b^{(k)} = b$), the centered vector $\tilde{\mu}^{(k)}$ simplifies to $b \cdot (2\pi_j^{(k)} - 1)$, which is perfectly collinear with $(u^{(k)} - \bar{u})$. As a result, the column space of $\mathbf{T}_\mathbf{II}$ is completely absorbed by the column space of $\mathbf{T}_\mathbf{I}$, collapsing the rank bound to:*

$$\text{rank}(\Sigma_{\text{off}}^{\text{pool}}) \leq K. \quad (21)$$

3 Third-Order Moment

Theorem 3.1 (Third-Order Tensor Decomposition with Explicit Scaling Factor). *Let $T_{j_1 j_2 j_3}$ represent the off-diagonal elements ($j_1 \neq j_2 \neq j_3$) of the third-order central moment tensor computed under the pooled mixture distribution:*

$$M_{j_1 j_2 j_3}^{(3)} = \mathbb{E}[(\tilde{v}_{j_1} - \mu_{j_1})(\tilde{v}_{j_2} - \mu_{j_2})(\tilde{v}_{j_3} - \mu_{j_3})]. \quad (22)$$

Under uniform class balance ($b^{(k)} = b$), $M^{(3)}$ admits a clean symmetric CP tensor decomposition:

$$M_{j_1 j_2 j_3}^{(3)} = \gamma(b) \sum_{k=1}^K \rho_k u_{j_1}^{(k)} u_{j_2}^{(k)} u_{j_3}^{(k)}, \quad (23)$$

where $\gamma(b) = \frac{-2b}{\sqrt{1-b^2}}$ is an explicit scalar multiplier relying entirely on the balance parameter b .

Important Inference: Derived directly from Eq 23, the second-order matrix $M^{(2)}$ and the third-order tensor $M^{(3)}$ share exactly identical decomposition directional factors $\{u^{(k)}\}$, with variations occurring solely within their respective scalar mixture weights. This structural coupling serves as the geometric foundation for parameters identifiability.

Proof. We evaluate the central moment by conditioning on the latent path (y, d_ℓ) :

$$M_{j_1 j_2 j_3}^{(3)} = \mathbb{E}_{d_\ell} \left[\mathbb{E}_{y,v} \left[\prod_{\ell=1}^3 (\tilde{v}_{j_\ell} - \mu_{j_\ell}) \mid d_\ell \right] \right]. \quad (24)$$

By the DCI assumption, when all three indices are distinct, the verifiers decouple conditionally given $(y, d_\ell = k)$:

$$\mathbb{E} \left[\prod_{\ell=1}^3 (\tilde{v}_{j_\ell} - \mu_{j_\ell}) \mid d_\ell = k \right] = \mathbb{E}_y \left[\prod_{\ell=1}^3 \mathbb{E}[\tilde{v}_{j_\ell} - \mu_{j_\ell} \mid y, d_\ell = k] \mid d_\ell = k \right]. \quad (25)$$

Under uniform class balance ($b^{(k)} = b$), the marginalized pooled mean reduces to $\mu_j = b(2\pi_j^{(k)} - 1)$. Using the identity $\mathbb{E}[\tilde{v}_j \mid y, k] = y(2\pi_j^{(k)} - 1)$, the centered component matches $\mathbb{E}[\tilde{v}_j - \mu_j \mid y, k] = (y - b)(2\pi_j^{(k)} - 1)$. Multiplying these terms gives:

$$\mathbb{E} \left[\prod_{\ell=1}^3 (\tilde{v}_{j_\ell} - \mu_{j_\ell}) \mid d_\ell = k \right] = \mathbb{E}_y [(y - b)^3 \mid d_\ell = k] \prod_{\ell=1}^3 (2\pi_{j_\ell}^{(k)} - 1). \quad (26)$$

We now evaluate the expected cubic difference over the Bernoulli label distribution $y \in \{-1, +1\}$. Expanding the term yields:

$$\mathbb{E}_y [y^3 - 3y^2b + 3yb^2 - b^3 \mid d_\ell = k] = \mathbb{E}_y [y - 3b + 3yb^2 - b^3 \mid d_\ell = k], \quad (27)$$

since $y^2 = 1$ and $y^3 = y$ deterministically in a binary space. Substituting $\mathbb{E}[y \mid k] = b$:

$$\mathbb{E}_y [(y - b)^3 \mid d_\ell = k] = b - 3b + 3b^3 - b^3 = -2b(1 - b^2). \quad (28)$$

Substituting this back into the outer expectation over regimes gives:

$$M^{(3)} = \sum_{k=1}^K \rho_k (-2b(1 - b^2)) (2\pi^{(k)} - 1) \otimes (2\pi^{(k)} - 1) \otimes (2\pi^{(k)} - 1). \quad (29)$$

To express this in terms of the rescaled parameters, we use the definition $(2\pi^{(k)} - 1) = u^{(k)} / \sqrt{1 - b^2}$. Substituting this into the tensor product yields:

$$M^{(3)} = \frac{-2b(1 - b^2)}{(1 - b^2)^{3/2}} \sum_{k=1}^K \rho_k u^{(k)} \otimes u^{(k)} \otimes u^{(k)} = \left(\frac{-2b}{\sqrt{1 - b^2}} \right) \sum_{k=1}^K \rho_k u^{(k)} \otimes u^{(k)} \otimes u^{(k)}. \quad (30)$$

Setting $\gamma(b) = \frac{-2b}{\sqrt{1 - b^2}}$ completes the proof. $\square$

4 Identifiability and Subspace Separation

Theorem 4.1 (Parameter Identifiability via Kruskal's Theorem). *Assume the following conditions hold:*

1. $\{u^{(1)}, \dots, u^{(K)}\}$ are linearly independent (Regime Separation),
2. Mixture probabilities are strictly positive ($\rho_k > 0$ for all k),
3. The number of available verifiers satisfies $m \geq 2K + 1$,
4. The majority of verifiers maintain a performance tier that is better-than-random within each specific regime.

Then the parameter set $\{\rho_k, u^{(k)}\}_{k=1}^K$ can be uniquely recovered (up to permutation) from the moments $(M^{(2)}, M^{(3)})$.

Remark 4.2 (Robustness to Rank Misspecification). *While our theoretical derivation assumes the number of regimes K is known, in practical applications, the eigengap $\lambda_K - \lambda_{K+1}$ of the diagonal-cleaned covariance matrix $\tilde{M}$ serves as a robust heuristic for selecting the optimal rank. Our spectral estimation framework exhibits inherent stability to small perturbations in the rank K : as U_K captures the dominant variance of the mixture components, minor over-estimation of K manifests as additional low-amplitude noise in the projected latent space, which is naturally attenuated by the subsequent k -means clustering and FUSE estimation.*

Proof. By Theorem 3.1, the off-diagonal third-order tensor is expressed as a symmetric rank- K CP decomposition. According to Kruskal's Uniqueness Theorem (1977), a three-way CP decomposition is unique up to permutation and rescal-

ing if the Kruskal ranks (krank) of the factor matrix $U = [u^{(1)}, \dots, u^{(K)}]$ satisfy:

$$3 \cdot \text{krank}(U) \geq 2K + 2. \quad (31)$$

Condition 1 (linear independence) directly ensures that U has full column rank, meaning $\text{krank}(U) = K$. Substituting this into Kruskal's inequality gives $3K \geq 2K + 2 \implies K \geq 2$. For $K = 1$, the model simplifies to the baseline single-regime FUSE model, where identification is straightforward. Since the global class balance b can be estimated from the pooled empirical mean, the scaling factor $\gamma(b)$ is uniquely determined, ensuring exact parameter recovery. $\square$

Lemma 4.3 (Subspace Separation of Regimes). *Under DCI assumptions and a rank- K decomposition of M_{signal} , the conditional expectation vectors $\mu^{(k)} := \mathbb{E}[\tilde{v} \mid d = k]$ are perfectly embedded within the subspace spanned by the top- K eigenvectors U_K of the diagonal-cleaned covariance matrix. Specifically, there exists a unique mapping such that $\mu^{(k)} = U_K \Lambda_K^{1/2} e_k$. Furthermore, these vectors satisfy a strict minimum separation boundary in the latent space:*

$$\|\mu^{(k)} - \mu^{(k')}\| \geq c > 0 \quad \text{for } k \neq k'. \quad (32)$$

Theorem 4.4 (k-means Clustering Consistency). *Let $z_i = U_K^\top \tilde{v}_i$ denote the lower-dimensional coordinates projected onto the empirical eigenbasis. Under the spectral eigengap condition and bounded noise, the centroids of the mixtures in the projected space match the latent regimes bijectively, and the k -means algorithm over $\{z_i\}$ consistently recovers the true regime partitions d_i as $NL \rightarrow \infty$.*

5 Spectral Estimation and Finite-Sample Robustness

Theorem 5.1 (Asymptotic Estimator Consistency). *Let $\hat{M}$ be empirical covariance. Then:*

$$\|\hat{M} - M\| = O_p(1/\sqrt{NL}). \quad (33)$$

Moreover, the perturbation angle of recovered subspaces vanishes asymptotically:

$$\sin \Theta(\hat{U}_K, U_K) \rightarrow 0. \quad (34)$$

Proof. By applying the matrix Bernstein inequality to the sum of independent bounded random vectors:

$$\hat{M} = \frac{1}{NL} \sum_{i=1}^{NL} \tilde{v}_i \tilde{v}_i^\top \implies \|\hat{M} - M\| = O_p(1/\sqrt{NL}).$$

Next, to bound the alignment error of the recovered singular subspace, we invoke the Davis–Kahan $\sin \Theta$ theorem. Let λ_K be the K -th largest eigenvalue and λ_{K+1} be the $(K+1)$ -th eigenvalue. The perturbation angle is bounded by:

$$\sin \Theta(\hat{U}_K, U_K) \leq \frac{\|\hat{M} - M\|}{\lambda_K - \lambda_{K+1}}.$$

Since $\text{rank}(M_{\text{signal}}) \leq K$, the noise-free boundary satisfies $\lambda_{K+1} = 0$. Condition 1 ensures that $\lambda_K > 0$, guaranteeing a strictly positive spectral gap. As the sample size grows ($NL \rightarrow \infty$), the numerator vanishes while the gap remains stable, proving consistency. $\square$

Theorem 5.2 (Finite-Sample Subspace Perturbation and Clustering Guarantee). *Let $\hat{M}$ be the empirical covariance matrix calculated from NL historical scores, and let $\tilde{M} = \hat{M} - \text{diag}(\hat{M})$. Let $\hat{U}_K$ and U_K be the top- K left singular subspaces of $\tilde{M}$ and M_{signal} , respectively. Assume the spectral gap condition holds, i.e., $\lambda_K(M_{\text{signal}}) \geq \Delta > 0$. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, we have:*

$$\sin \Theta(\hat{U}_K, U_K) \leq \frac{C_1}{\Delta} \sqrt{\frac{m \log(m/\delta)}{NL}}, \quad (35)$$

where C_1 is a universal constant. Furthermore, if the minimum regime separation c satisfies $c > \frac{4C_1}{\Delta} \sqrt{\frac{m \log(m/\delta)}{NL}}$, the misclassification rate of the k -means algorithm on the projected coordinates $z_i = \hat{U}_K^\top \tilde{v}_i$ scales boundedly as:

$$\mathbb{P}(\hat{d}_i \neq d_i) = O\left(\frac{m \cdot K}{\Delta^2 \cdot c^2 \cdot NL}\right) \rightarrow 0 \quad \text{as } NL \rightarrow \infty. \quad (36)$$

Theorem 5.3 (FUSE Misspecification Collapse under Regime Heterogeneity). *Suppose the true data-generating mechanism follows the DCI framework with $K \geq 2$ dynamic regimes, where $\text{rank}(M_{\text{signal}}) = K$. Let $\hat{u}_{\text{FUSE}}$ be the reliability vector estimated by the standard baseline FUSE model (which flatly enforces a rank-1 assumption). If there exist at least two regimes $k \neq k'$ such that their structural reliability vectors are non-collinear ($u^{(k)} \not\propto u^{(k')}$), then even in the asymptotic limit as the sample size approaches infinity ($NL \rightarrow \infty$), the FUSE estimator suffers from an irreducible systemic bias:*

$$\lim_{NL \rightarrow \infty} \|\hat{u}_{\text{FUSE}} - \mathbb{E}_d[u^{(d)}]\|_2^2 \geq \sum_{k=1}^K \rho_k \|u^{(k)} - \mathcal{P}_{u_{\text{FUSE}}}(u^{(k)})\|_2^2 \geq \epsilon > 0, \quad (37)$$

where $\mathcal{P}_{u_{\text{FUSE}}}$ denotes the orthogonal projection operator onto the space spanned by $\hat{u}_{\text{FUSE}}$, and ϵ is a strictly positive constant proportional to the angular variance of the across-regime profiles.

6 Algorithm

Algorithm 1 DCI-FUSE (Regime-Agnostic Decomposition with Diagonal Cleaning)

- 1: Input: verifier outputs $\{v_{\ell ij}\}$
 - 2: Compute empirical covariance matrix $\hat{M}$ across all NL pooled responses.
 - 3: Remove non-informative diagonal variance noise: $\tilde{M} = \hat{M} - \text{diag}(\hat{M})$.
 - 4: Eigen-decomposition on the diagonal-cleaned matrix: $\tilde{M} = \hat{U} \hat{\Lambda} \hat{U}^\top$.
 - 5: Keep top K eigenvectors $\hat{U}_K$ to isolate the mixture subspace.
 - 6: Project responses into the lower-dimensional latent space: $z_i = \hat{U}_K^\top \tilde{v}_i$.
 - 7: Execute k -means clustering on the projected coordinates $\{z_i\}$ to partition responses into K latent regimes $\{\hat{d}_i\}$.
 - 8: Estimate cluster-wise reliability vectors $\{\hat{u}^{(k)}\}$ and mixing weights $\{\hat{\rho}_k\}$ by applying standard FUSE to each isolated regime cluster.
 - 9: Output: Discovered regime parameters $\{\hat{u}^{(k)}, \hat{\rho}_k\}$.
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